• DRIVEABILITY—EXCESSIVE SULFUR SMELL FROM EXHAUST • EXHAUST—EXCESSIVE SULFUR SMELL FROM EXHAUST	Article No. 02-21-6

FORD: 2001-2002 EXPLORER SPORT
2002 EXPLORER SPORT TRAC
2002-2003 EXPLORER

MERCURY: 2002-2003 MOUNTAINEER

Article **02-16-6** is being republished in its entirety to update the Service Procedure and model years.

ISSUE

Some vehicles may exhibit excessive sulfur smell from the exhaust under certain drive modes. This may be due to the highly active state of a new catalyst, calibration factors, underbody catalyst temperature, and amount of sulfur present in the fuel.

ACTION

Identify year, model, and build date to identify appropriate service procedure. For applications that require installing exhaust wrap, use good quality stainless steel worm-drive hose clamps for best longevity, similar to what would commonly be used for exhaust heat shield rattle issues.

SERVICE PROCEDURE

2002-2003 4 DOOR EXPLORER/MOUNTAINEER 4.0L

Vehicles Built 1/14/02 Through 12/02/2002

At latest level for engine calibration.

1. Install Exhaust wrap on the pipes between the light-off and the underbody catalyst referring to the instruction sheet in the kit.
2. Refer to Drive Sequence heading to purge stored sulfur from catalyst.
3. See Contributing Factors heading for additional information.

NOTE
CONVERTER REPLACEMENT OR RECALIBRATION IS NOT NECESSARY.

Vehicles Built Before 1/14/02

1. Verify that WDS is at the latest level, B21.0 or higher.

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2. Reprogram the PCM with the latest calibration.
3. Install Exhaust wrap on the pipes between the light-off and the underbody catalyst referring to the instruction sheet in the kit.
4. Refer to Drive Sequence heading to purge stored sulfur from catalyst.
5. See Contributing Factors Heading for additional information.

NOTE
DO NOT REPLACE THE CONVERTER.

2002-2003 4 DOOR EXPLORER/MOUNTAINEER 4.6L

Vehicles Built Prior To 12/02/2002

1. Install Exhaust wrap on the pipes between the light-off and the underbody catalyst referring to the instruction sheet in the kit.
2. Refer to Drive Sequence heading to purge stored sulfur from catalyst.
3. See Contributing Factors Heading for additional information.

NOTE
CONVERTER REPLACEMENT OR RECALIBRATION IS NOT NECESSARY.

2001-2002 2 DOOR EXPLORER SPORT 4.0L

7/24/00 Through 3/4/02 With 5R55E Transmission

1. Refer to 2001/2002 Explorer Sport/Sport Trac Workshop Manual, Section 309-00. Remove the old catalytic converter and install the revised converter 2L2Z-5E212-DA.
2. Verify that WDS is at the latest level, B21.0 or higher.
3. Reprogram the PCM with the latest calibration only.
4. See Contributing Factors Heading for additional information.

NOTE
DO NOT REPLACE CONVERTER MORE THAN ONCE.

2002 2 DOOR EXPLORER SPORT/SPORT TRAC 4.0L

Vehicles Built After 3/27/02

At latest level for engine calibration and converter, no service action will improve the situation.

1. See Contributing Factors Heading for additional information.

3/4/02 Through 3/27/02

1. Refer to 2002 Explorer Sport/Sport Trac Workshop Manual, Section 309-00. Remove the old catalytic converter and install the revised converter 2L2Z-5E212-AB only.
2. See Contributing Factors Heading for additional information.

NOTE
DO NOT REPLACE CONVERTER MORE THAN ONCE OR RECALIBRATE.

2002 2 DOOR EXPLORER SPORT/SPORT TRAC 4.0L M5R4 MANUAL TRANSMISSION

11/2/01 Through 3/4/02 With Manual Transmission

1. Refer to 2002 Explorer Sport/Sport Trac Workshop Manual, Section 309-00. Remove the old catalytic converter and install the revised converter 2L2Z-5E212-DA only.
2. See Contributing Factors Heading for additional information.

NOTE
DO NOT REPLACE CONVERTER MORE THAN ONCE OR RECALIBRATE.

2002 SPORT TRAC 4.0L 5R55E TRANSMISSION

11/2/01 Through 3/4/02 With 5R55E Transmission

1. Refer to 2002 Explorer Sport/Sport Trac Workshop Manual, Section 309-00. Remove the old catalytic converter and install the revised converter 2L2Z-5E212-DA.
2. Verify that WDS is at the latest level, B21.0 or higher.
3. Reprogram the PCM with the latest calibration.
4. See Contributing Factors Heading for additional information.

NOTE
DO NOT REPLACE CONVERTER MORE THAN ONCE.

DRIVE SEQUENCE AFTER REPAIRS COMPLETED

Check engine oil and coolant levels. Verify both are at specified levels before driving vehicle. Perform highway driving at normal highway speeds for 30 miles. The optimum engine speed for this drive sequence is 3,000 to 4,500 rpm, and this can be accomplished by manually selecting the transmission gear state. The goal of the drive procedure is to maintain a constant high temperature level in the exhaust system through the use of increased engine speed. The combination of increased temperature and driving will provide a sulfur cleaning environment for the catalyst system.

CONTRIBUTING FACTORS TO SULFUR SMELL

Factors outside of engineering control may lead to a sulfur odor. A large contributing factor is the amount of sulfur in the fuel. Replacing a Catalytic Converter multiple times will not alleviate the sulfur odor. Several catalyst replacements on the same vehicle may actually increase customer sensitivity to an exhaust odor concern. Based on some driving habits the converter does not get hot enough to burn out stored sulfur residue and can produce a sulfur odor. For example:

- Extended Idle - Increase sulfur odor
- Stop and go driving - Increase sulfur odor
- Trailer Towing - Decrease sulfur odor
- Heavy Loads - Decrease sulfur odor
- Short trips - Increase sulfur odor
- Abrupt accel/decel - Decrease sulfur odor
- Hotter driving conditions - Decrease sulfur odor
- Hilly terrain - Decrease sulfur odor
- Canceling Overdrive - Decrease sulfur odor

Catalysts with 5,000-10,000 Miles (8,000-16,000 Kilometers) are likely to have a sulfur smell due to the highly active state of the new catalyst. Refer to symptom chart #22 of PCED for appropriate year. Additives in fuels (Ethanol, Methanol, ETC.) can cause sulfur odor, suggest using a different fuel source. Geographic variations on sulfur odor intensity are possible due to different refineries supplying fuel to different areas of the country. California regulated fuel will exhibit less sulfur odor.

Experiment with different fuel brands. Some contain more sulfur than others, which can lead to an increase in sulfur odor from the converter. Sulfur odors are an aromatic phenomenon and do not cause any durability or safety concerns with vehicle operation.

To summarize for the customer, explain that sulfur is in the fuel used and the catalyst system on their vehicle as a result of the fuel and driving environment which creates a condition where sulfur has been stored in their exhaust system. The service action performed allows the sulfur to pass through the exhaust system and to clean the existing sulfur deposits. During the initial cleansing of the catalyst it is expected that a sulfur odor may still be present, and this odor may change from the 'rotten egg smell' to that of a 'burning match'. The odor will cease with time.

Article No. 02-21-6 Cont'd.

PCM CALIBRATION INFORMATION							
Application	Old Part Number (-12A650-)	Old Tear Tag	New Part Number (-12A650-)	New Tear Tag	Old Calibration	New Calibration	NGS/WDS Qualifier
4.0L	1L2F-AGM	LAE1	1U7Z-BZA	ZPZ0	1U51AF0B15	1U51AF0B15	WDS Release B21.1 or Later
4.0L	1L2F-ACM	CNK1	1U7Z-CBA	CVF0	1U51BG0A15	1U51BG0A15	WDS Release B21.1 or Later
4.0L	1L2A-BL	SIM0	1U7Z-CTA	TFJ0	1U51AG0M15	1U51AG0M15	WDS Release B21.1 or Later
4.0L	1L2A-AEB	AUE1	1U7Z-CYA	YTD0	1U51AF0B21	1U51AF0B21	WDS Release B21.1 or Later
4.0L	1L2A-AFB	XPB1	1U7Z-CZA	ZAR0	1U51BG0A21	1U51BG0A21	WDS Release B21.1 or Later
4.0L	1L2A-AGB	EAB1	1U7Z-DFA	FMM0	1U51AG0M21	1U51AG0M21	WDS Release B21.1 or Later
4.0L	1L2A-AHB	KVO1	1U7Z-DGA	GBH0	1U51A40E21	1U51A40E21	WDS Release B21.1 or Later
2 Door 4.0L	1U7A-CGA	GZM0	1U7Z-CGB	GZM1	1U71AG0510	1U71AG0510	WDS Release B21.1 or Later
2 Door 4.0L	1U7A-CJA	JJU0	1U7Z-CJB	JJU1	1U71AG0510	1U71AG0510	WDS Release B21.1 or Later
2 Door 4.0L	1U7A-CHA	HBC0	1U7Z-CHB	HBC1	1U72AG0506	1U72AG0506	WDS Release B21.1 or Later
2 Door 4.0L	1U7A-CKA	KNW0	1U7Z-CKB	KNW1	1U72AG0506	1U72AG0506	WDS Release B21.1 or Later
4.0L	1U7A-CMA	MQS0	1U7Z-CMB	MQS1	1S12A40506	1S12A40506	WDS Release B21.1 or Later
4.0L	1U7A-CLA	LFA0	1U7Z-CLB	LFA1	1S11A40510	1S11A40510	WDS Release B21.1 or Later

Obtain an Authorized Modifications Decal (FPS 8262 - obtainable through DOES II, 25/pkg) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.

Ford AUTHORIZED MODIFICATIONS

THE FOLLOWING MODIFICATIONS HAVE BEEN MADE:

Reprogrammed Powertrain Control Module (PCM) Per TSB 02-21-6

THESE MODIFICATIONS HAVE BEEN APPROVED, AS APPROPRIATE, BY EPA AND CARB.

DEALER NUMBER: **DATE:**

CHANGE AUTHORITY:

FPS 8262 9/78 FORD MOTOR COMPANY PRINTED IN U.S.A.

PART NUMBER	PART NAME
2L2Z-4K277-AA	Exhaust Wrap Kit
2L2Z-5E212-DA	4.0L 5R55E Catalyst
2L2Z-5E212-AB	4.0L Catalyst

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 02-16-6
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
022106A	Install Exhaust Wrap And Perform Drive Sequence. 2002-2003 Explorer/Mountaineer 4.0L Vehicles Built 1-14-02 Through 12-02-02, 4.6L Built Prior To 12-02-02	1.4 Hrs.
022106B	Install Exhaust Wrap, Reprogram Powertrain Control Module And Perform Drive Sequence. 2002-2003 Explorer/Mountaineer 4.0L Built Prior To 1-14-02	1.6 Hrs.
022106C	Replace Catalytic Converter And Reprogram Powertrain Control Module. 2001-2002 Explorer Sport Vehicles Built 7-24-00 Through 3-4-02, 2002 Sport Trac 11-02-01 Through 3-4-02	0.8 Hr.
022106D	Replace Catalytic Converter Only. 2002 Explorer Sport/Sport Trac 3-4-02 Through 3-27-02 With Automatic Transmission, 11-02-01 Through 3-4-02 With Manual Transmission	0.6 Hr.

DEALER CODING

BASIC PART NO.	CONDITION CODE
5F250	42

OASIS CODES: 403000, 499000

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.